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Total No. of Questions : 8] [Total No. of Printed Pages : 3

Paper Code : 21324

F–424

B.C.A. (Vth Semester) Examination, 2019–20

(New Course)

SOFTWARE ENGINEERING

Paper–BCA–504–N

Time : 3 Hours] [Maximum Marks : 70

Note :– Attempt any *five* questions. All questions carry equal marks. Symbols used have their usual meaning.

1. (a) Define software engineering. Explain various software myths and applications.

   (b) Explain the various stages of software development in detail.

2. (a) What is E-R diagram ? Write down the various notations used in ER diagram. What are the various purposes of ER diagram in software development ?

   (b) Describe the major components (with their corresponding symbol) of DFD and also explain the various types of DFD with example.

3. (a) Explain the function oriented and object oriented approaches for software design.

   (b) What is requirement analysis ? Briefly explain different types of requirement problems that should be identified and resolved during requirement analysis activity.

4. (a) What is cohesion and coupling ? Write the differences between them.

   (b) Explain the COCOMO model in detail with the help of various equations. Compare the basic COCOMO model with the detailed COCOMO model.

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5. (a) What is software testing ? Explain its various objectives and principles.

   (b) Explain the functional testing and differentiate it with structural testing.

6. (a) Describe the role of quality control and quality assurance in software quality.

   (b) Describe the SEI CMM model with its different levels in detail.

7. (a) Explain reverse engineering in detail. Write the difference between reverse engineering and re-engineering.

   (b) Explain 4GL tools in detail.

8. Explain any *four* of the following :

   (i) SRS with characteristics

   (ii) Alpha & Beta Testing

   (iii) Verification Vs. Validation

   (iv) ISO 9001

   (v) CASE Tools

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